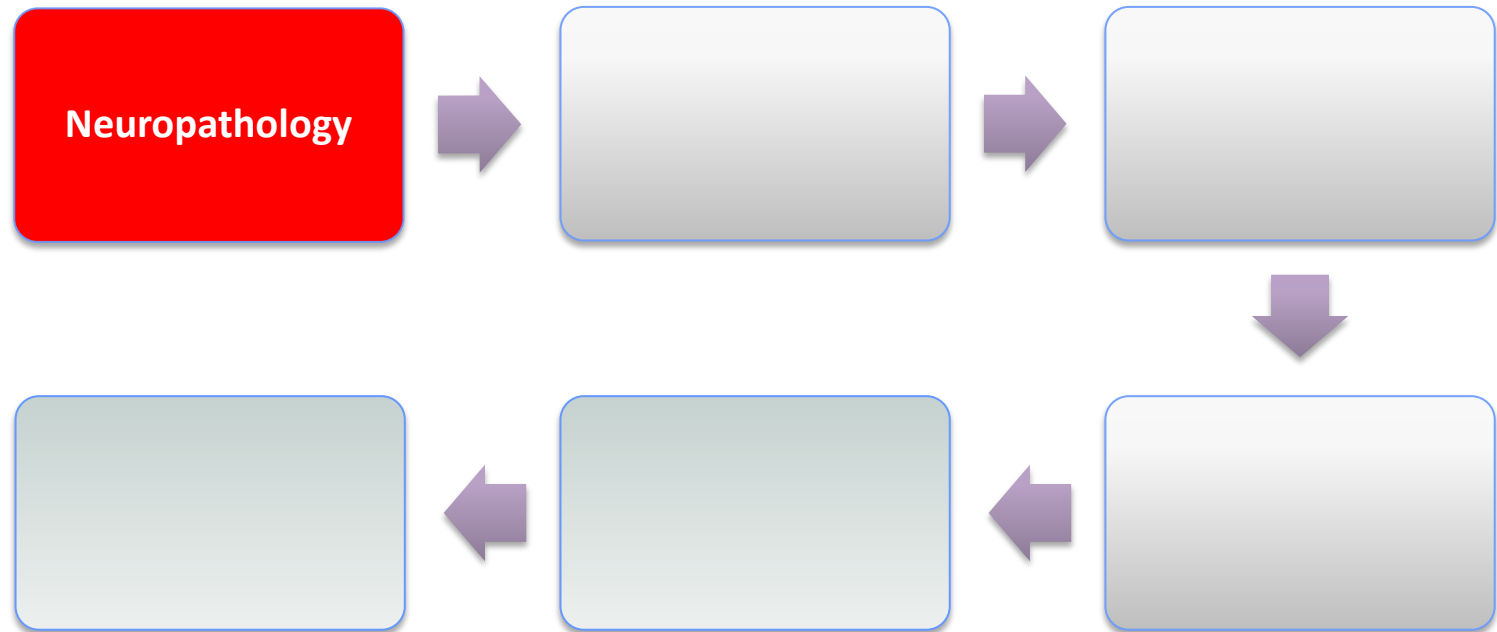


Dementia

Trends and Updates





Memory
Loss

Cognitive
Decline

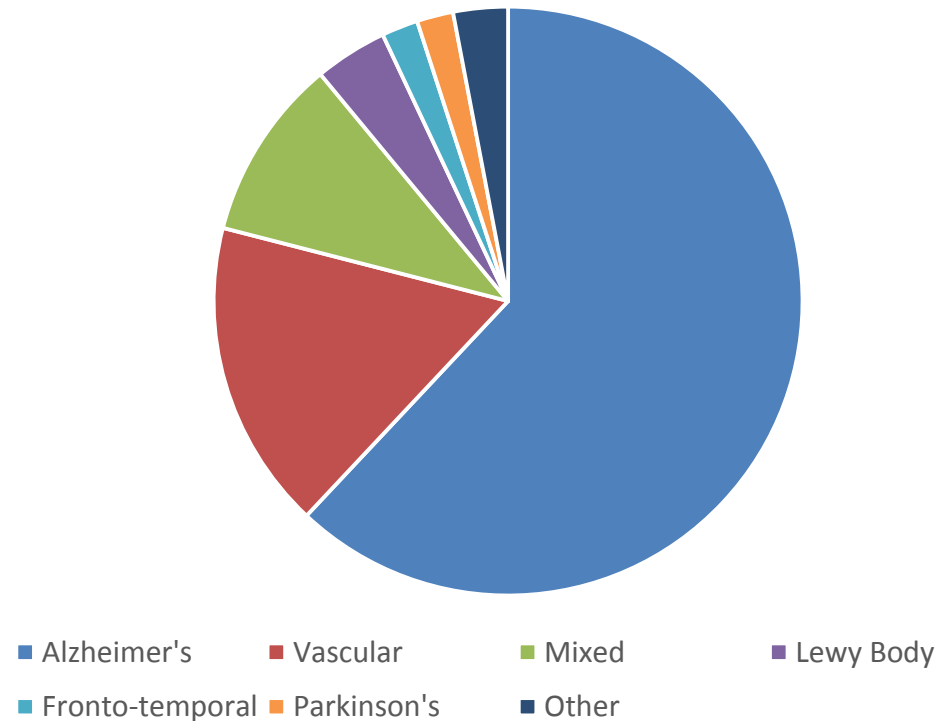
Problem-
solving

Language

Progressive

Incurable

Prevalence of types of dementia



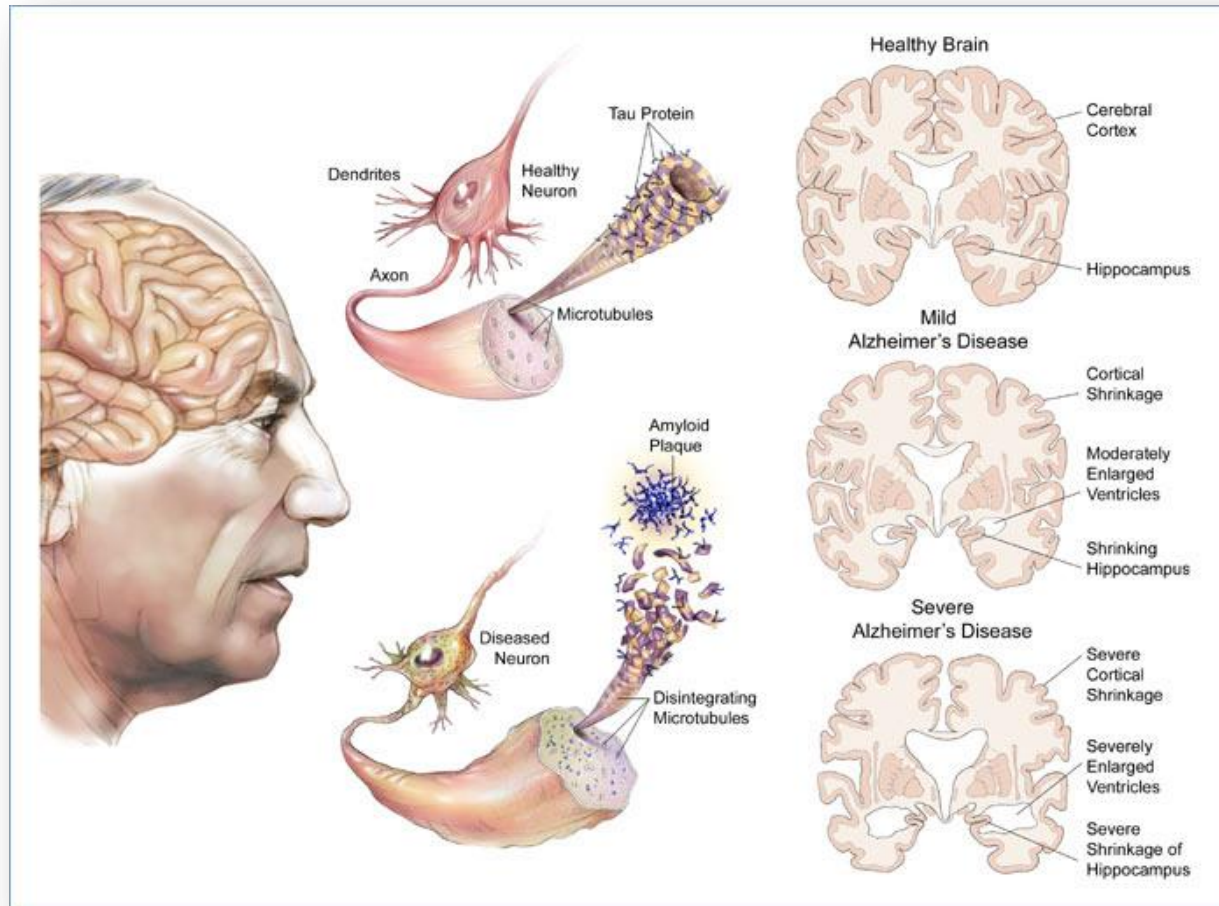
Alzheimer's Disease

- Toxic build-up of protein in the forms of plaques and tangles
- Both sporadic and familial
- Early symptoms involve memory loss and confusion
- Possible association with classic CVD risk factors

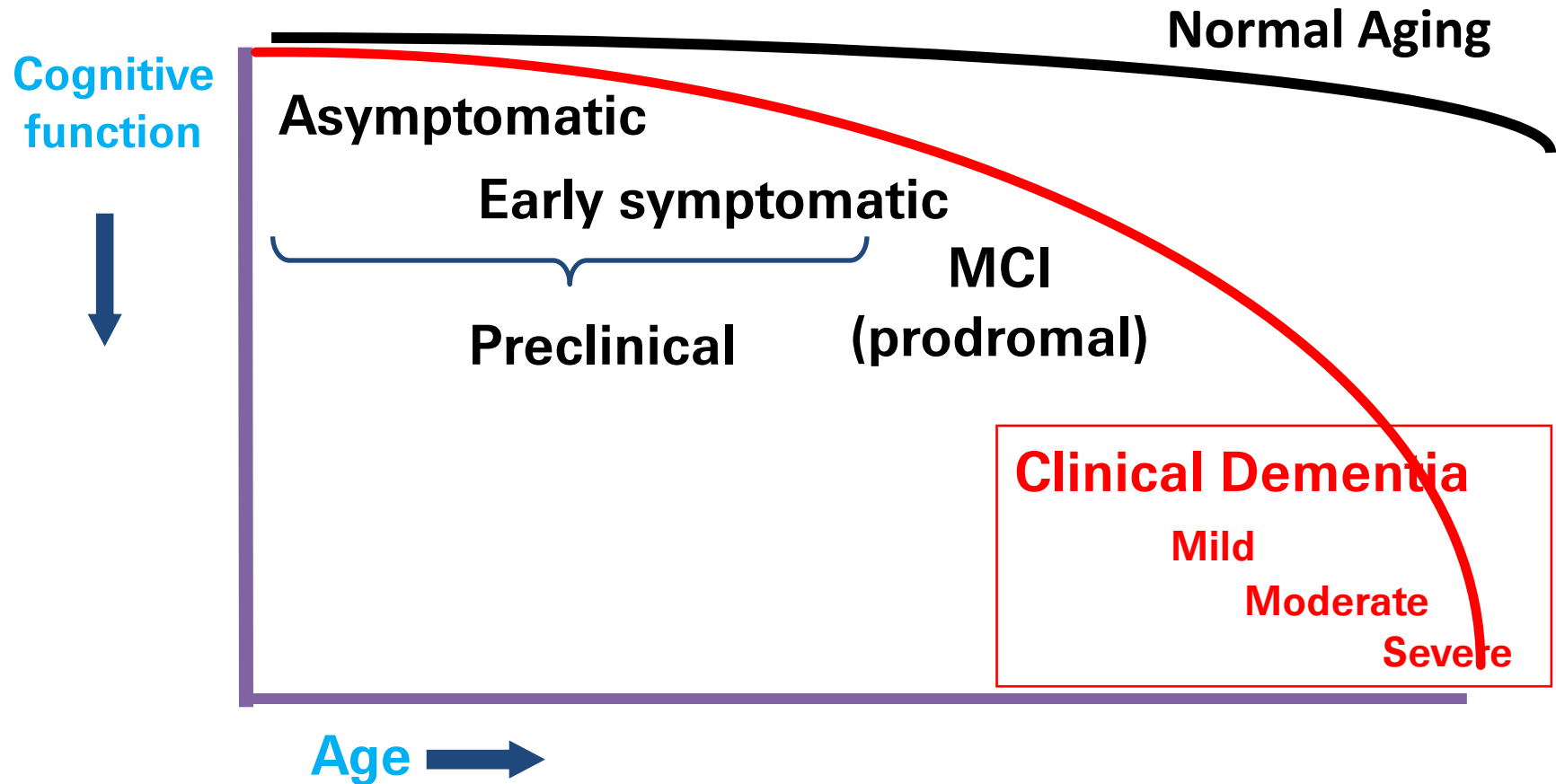
Vascular Dementia

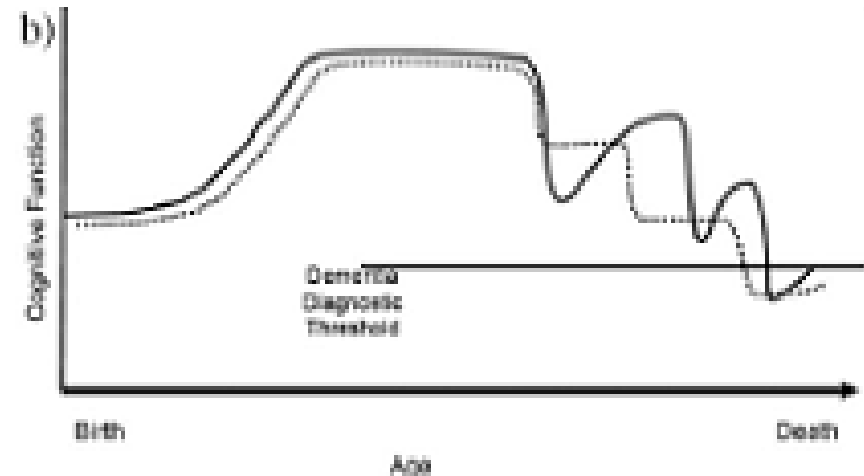
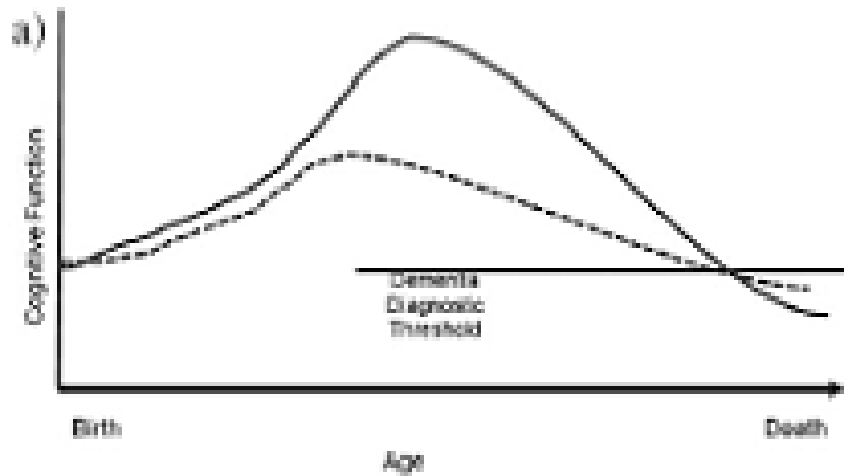
- Reduced blood supply due to diseased vessels
- Stroke-related or sub-cortical
- Early symptoms may not involve memory loss as in AD with stepped progression
- Associated with classic CVD risk factors

Neuro-pathology of Alzheimer's Disease

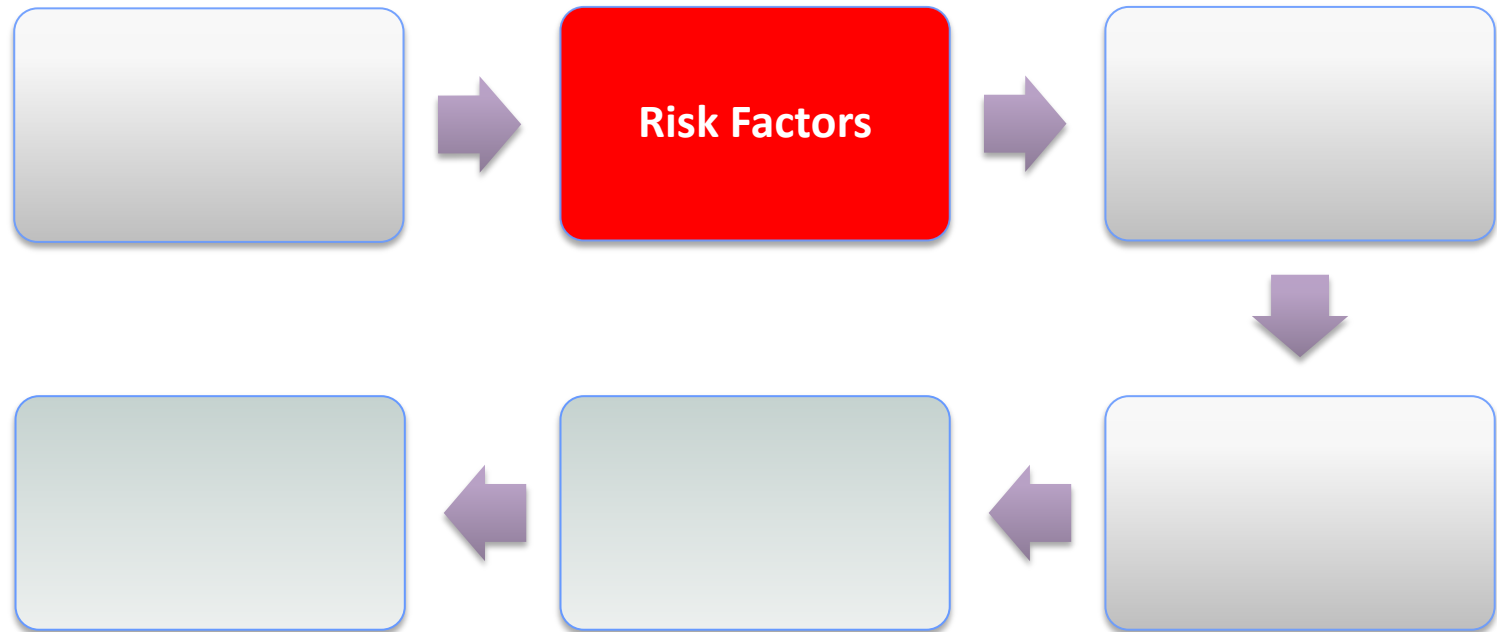


Alzheimer's Disease Trajectory





Alternative cognitive trajectories across the life-course, resulting in meeting threshold criteria at the same age. (a) Longer period of cognitive development but faster rate of age-related decline. (b) More severe cognitive decrements after stroke, but greater neurologic recovery.



Age

Doubling of risk every 5 years after the age of 65

Increased incidence of CVD

General aging of nerve cells, DNA and immune system

Genetics

ApoE

- Fat and protein transport system

APP, PSEN1/2

- Protein regulation (amyloid)
- Early onset

Cardiovascular disease risk factors

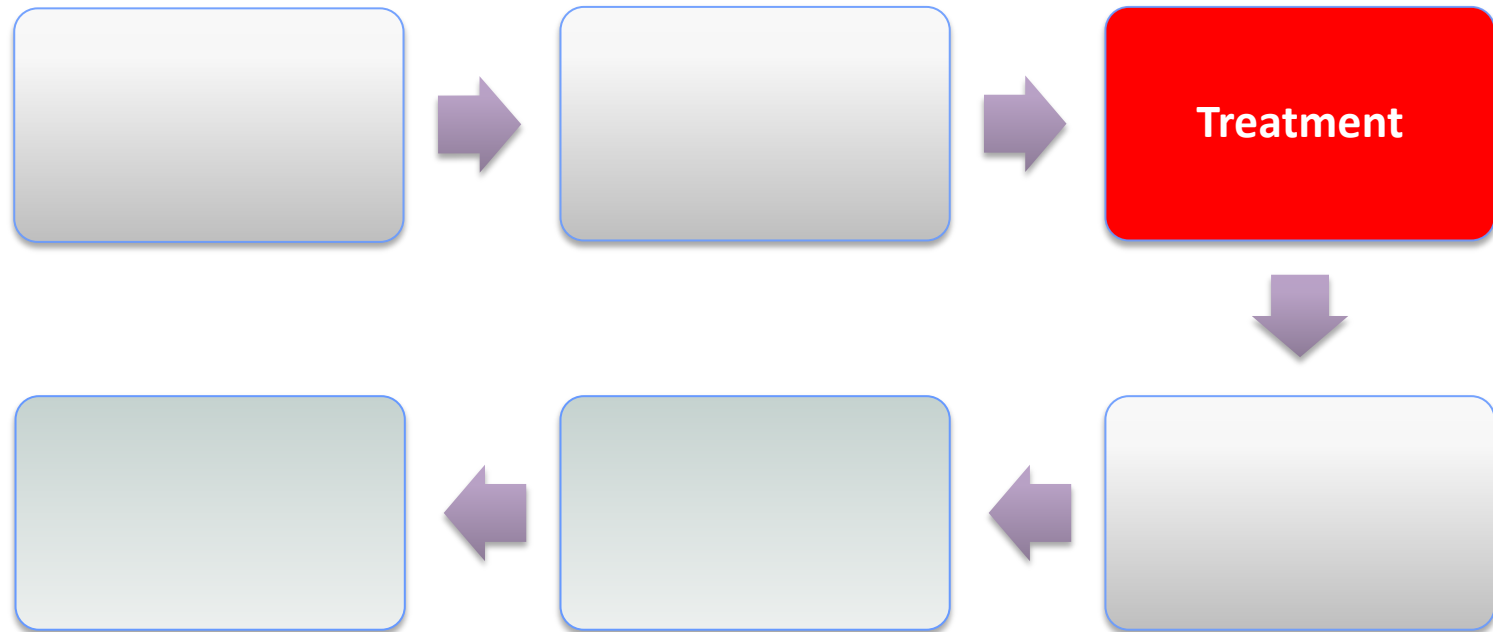
Stroke

VaD 

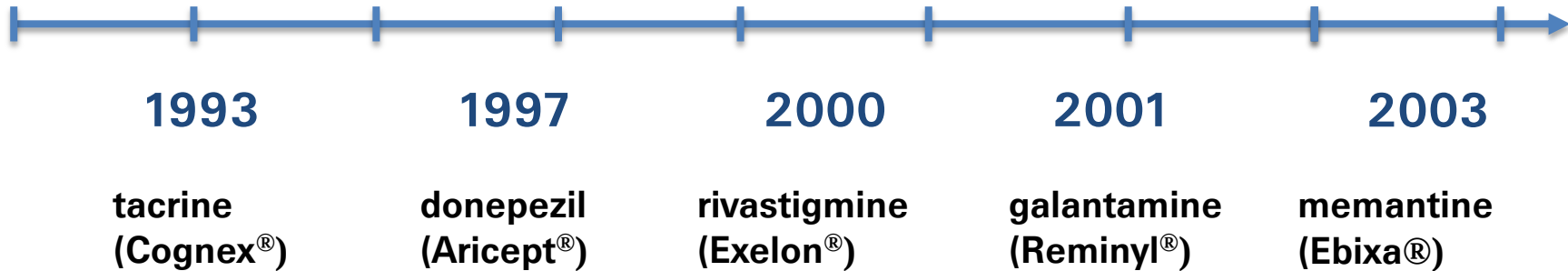
AF

CHD

Heart Failure



Current Treatment Approaches for AD



Issue date: [March 2011](#)

NHS
*National Institute for
Health and Clinical Excellence*

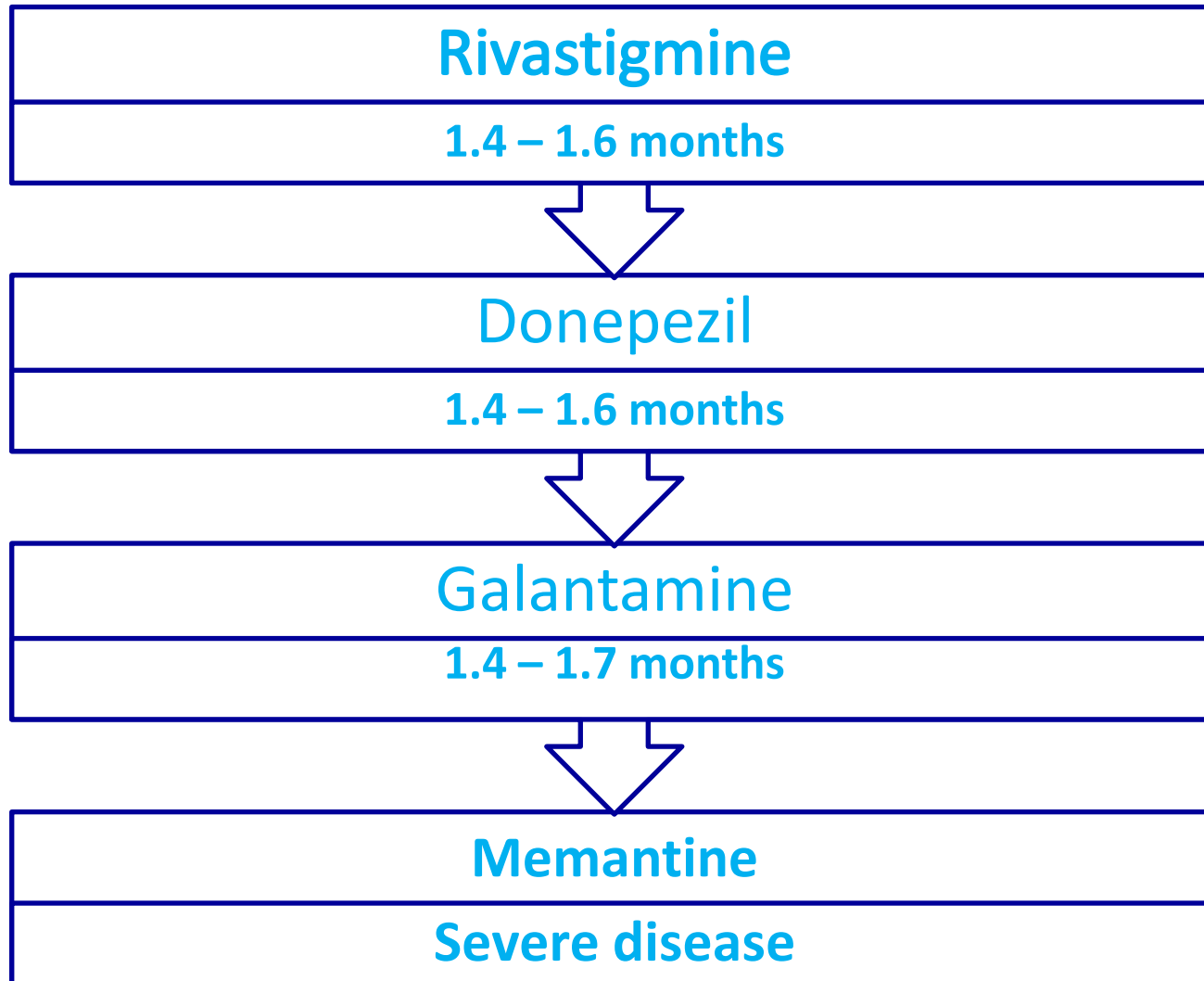
Quick reference guide

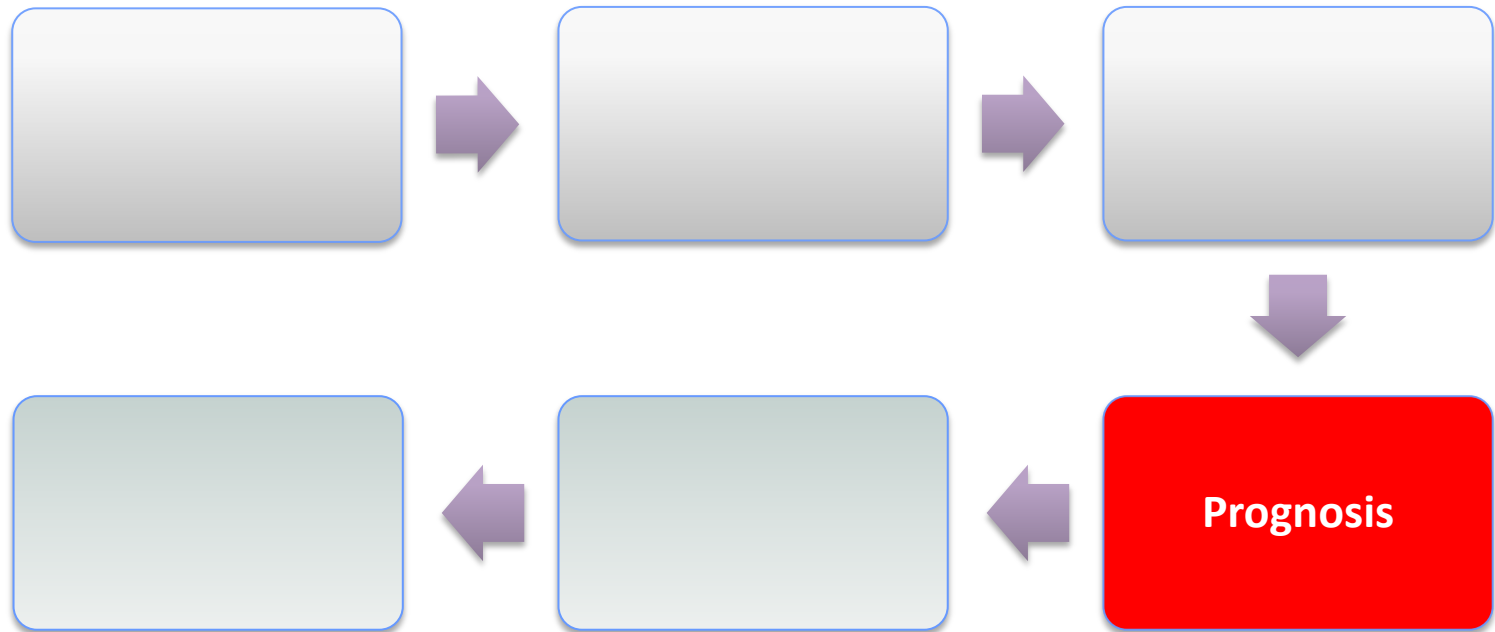
**Donepezil, galantamine, rivastigmine and
memantine for the treatment of Alzheimer's
disease (review)**

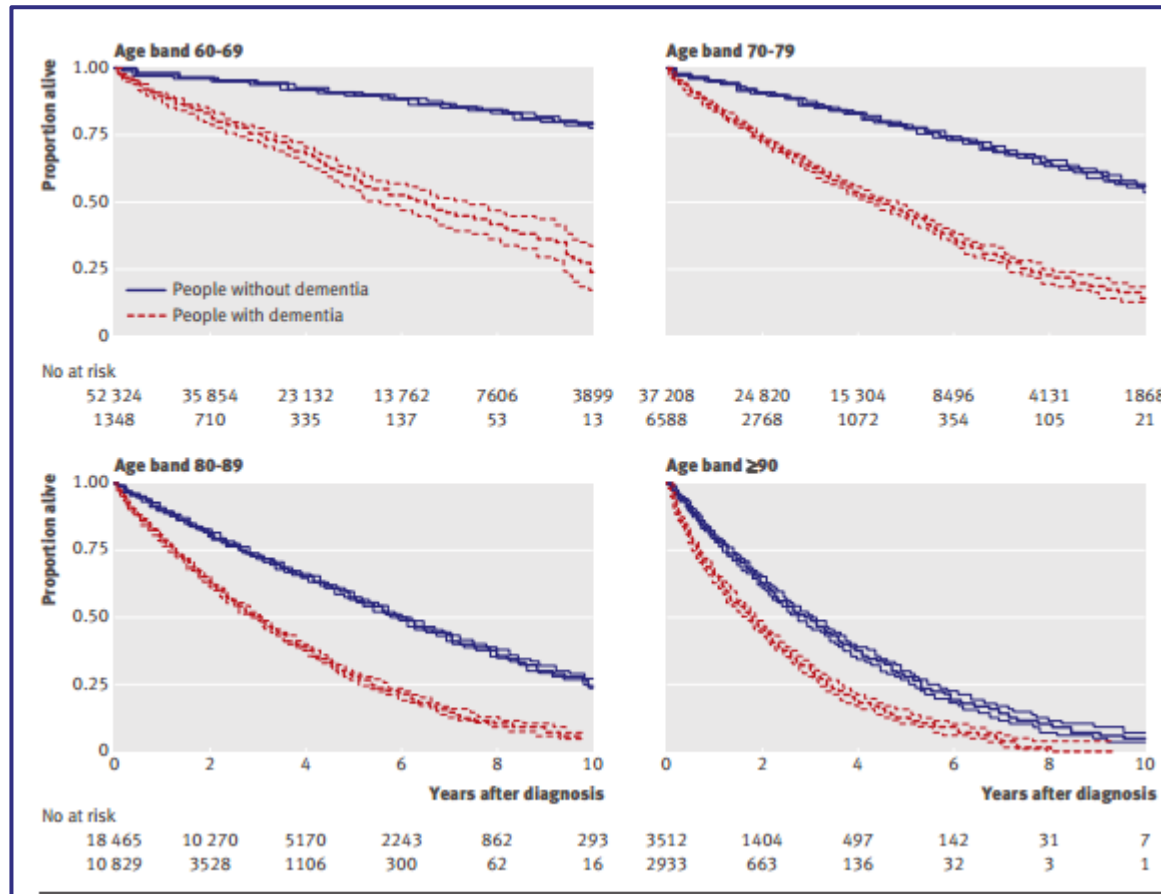
NOTE: This guidance replaces NICE technology appraisal guidance 111 issued in November 2006 (amended September 2007, August 2009).
The review and re-appraisal of donepezil, galantamine, rivastigmine and memantine for the treatment of Alzheimer's disease has resulted in a change in the guidance. Specifically:

- donepezil, galantamine and rivastigmine are now recommended as options for managing mild as well as moderate Alzheimer's disease, **and**
- memantine is now recommended as an option for managing moderate Alzheimer's disease for people who cannot take AChE inhibitors, and as an option for managing severe Alzheimer's disease.

Pharmaceutical Approaches

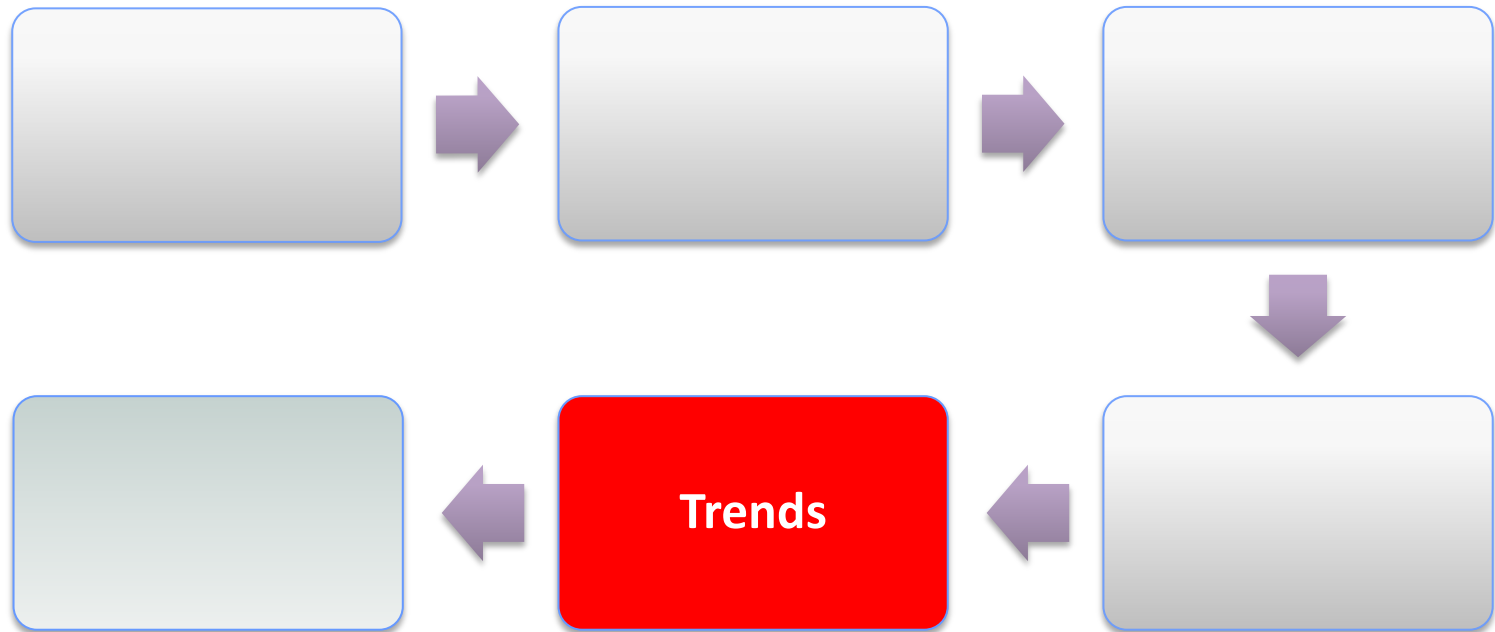




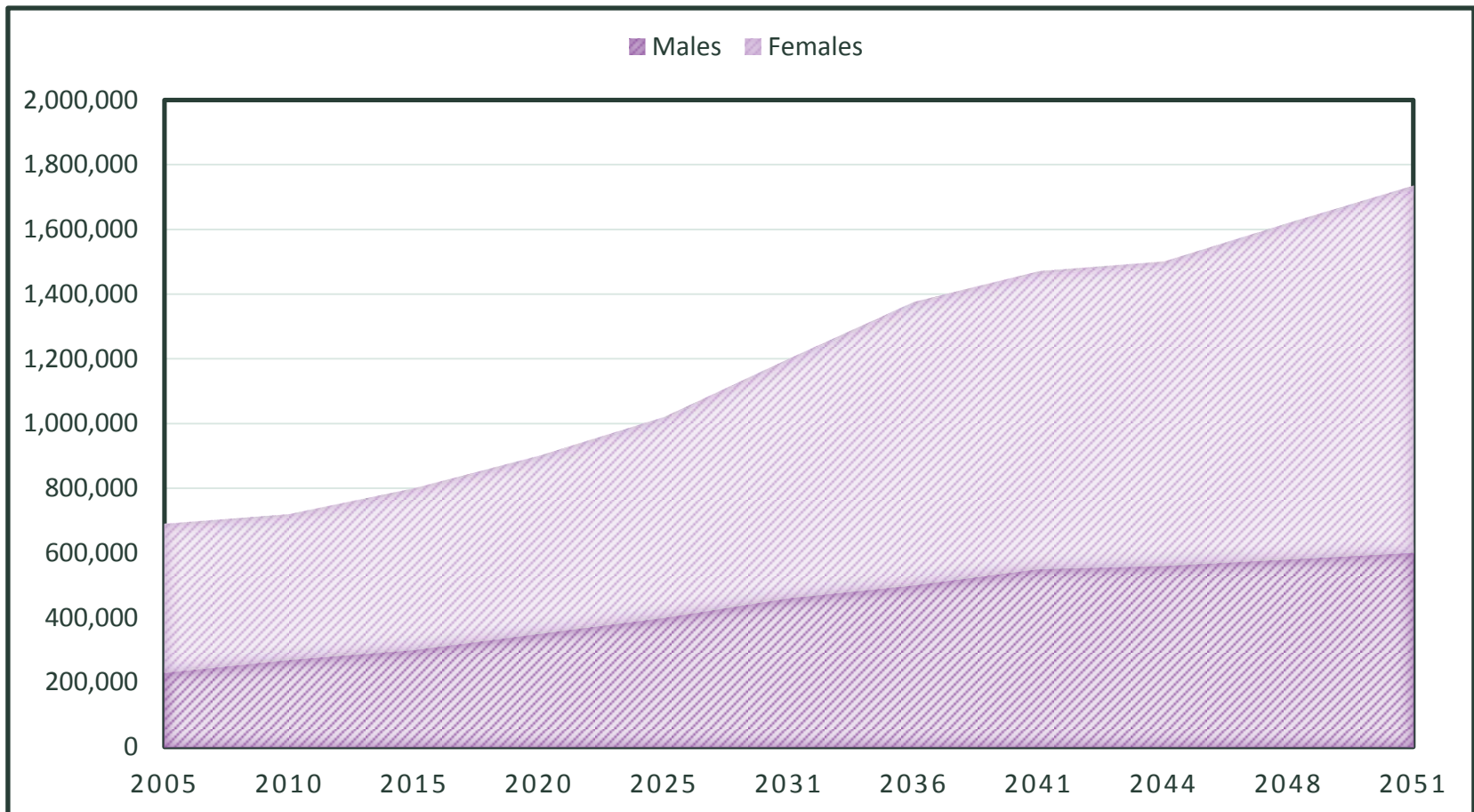


Rait et al (2010)
Survival of
people with
clinical diagnosis
of dementia in
primary care:
cohort study BMJ
2010;341:c3584

Kaplan-Meier survival curves for people with and without dementia



Future Prevalence – by gender – UK – calculated in 2007



Source; Alzheimer's Society (2014) Dementia UK

National Long-term care survey (US)

- Decline in prevalence of severe cognitive impairment 1982-1999, 5.7% to 2.9%

US Health and Retirement Study

- Decline in cognitive impairment between 1993-2002 from 12.2% to 8.7%

UK Cognitive Function and Ageing Study

- Decline in dementia prevalence rates from 8.3% to 6.5% between 1991 and 2011

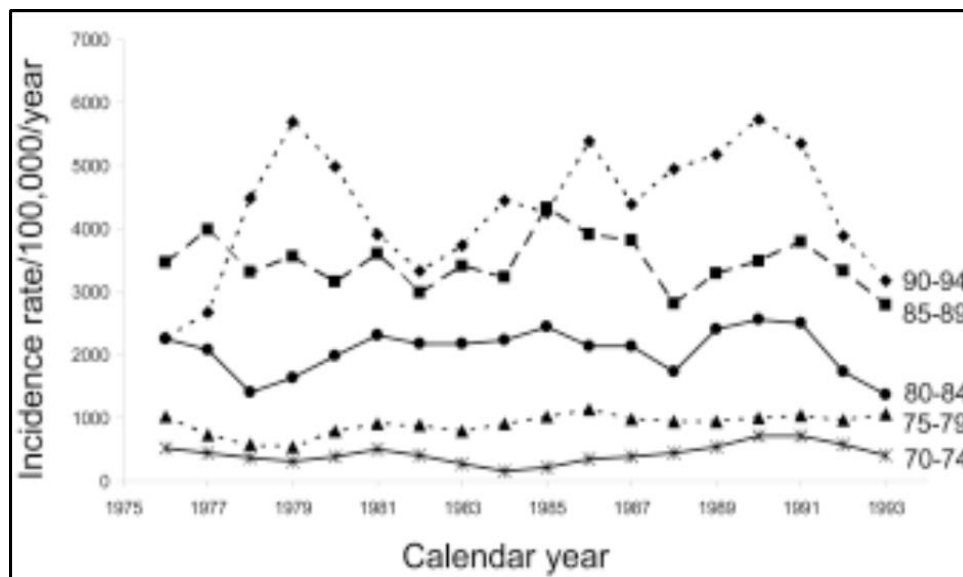
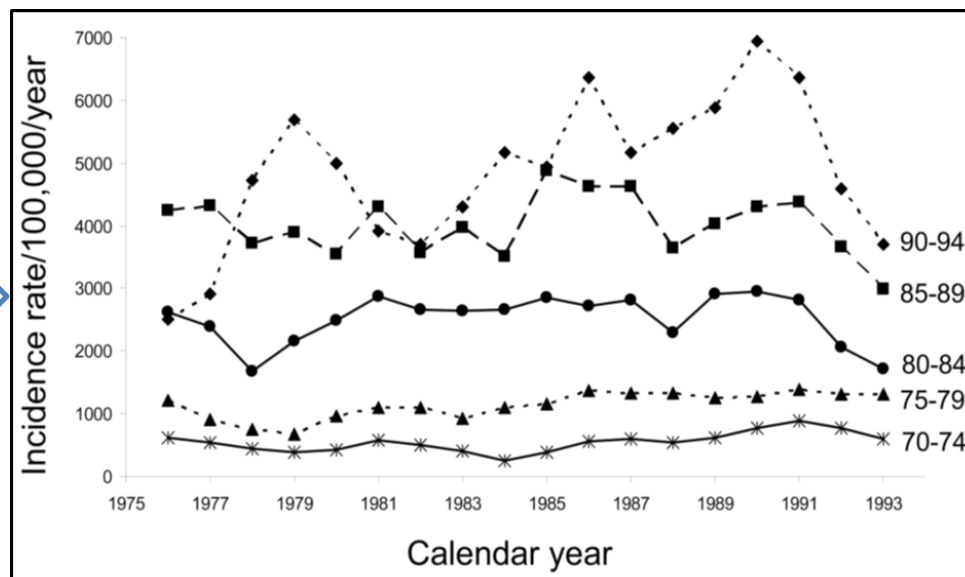
Rotterdam Study

- Observed decrease in dementia incidence between 1990 and 2000 – increase in treatment for cardiovascular risk factors

Framingham (US)

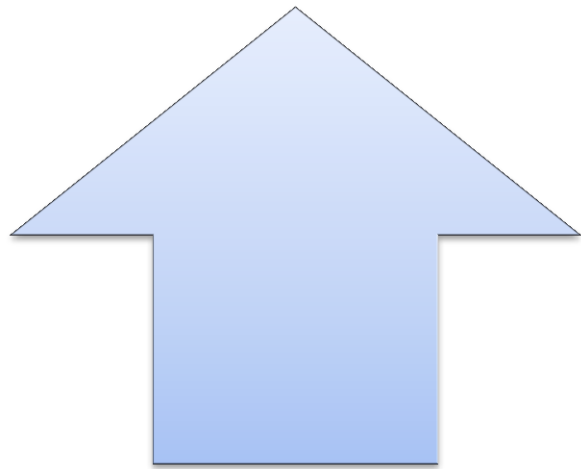
- Declines in dementia incidence across past 30 years
- Greatest decrease seen in women
- Associated with education, better management of CVD risks

Time trends in age-specific incidence rates of dementia in men and women combined from 1975 through 1994 (moving 3-year average incidence rates per 100,000 person-years): Rochester, Minnesota

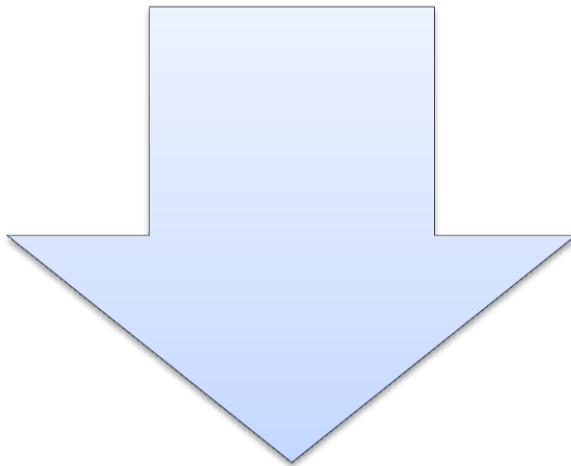


Time trends in age-specific incidence rates of Alzheimer's disease in men and women combined from 1975 through 1994 (moving 3-year average incidence rates per 100,000 person-years): Rochester, Minnesota.

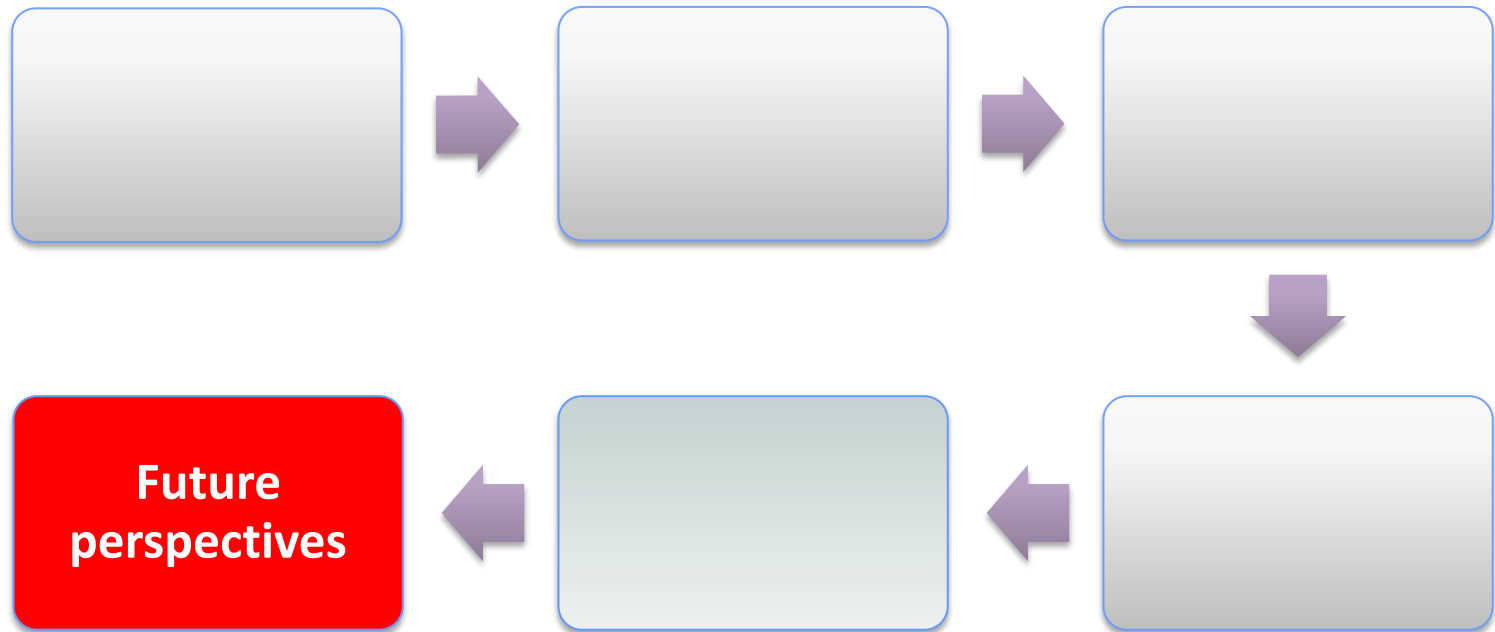
Risk factor trends



Obesity
Diabetes



Smoking



Aducanumab

- Phase Ib Study
- 166 patients 54 weeks with mild AD
- Does- and time-dependent clearance of amyloid and slowing of cognitive decline
- Fast track to phase III

Brain-training

- Study of participants with FH of AD
- Protective effects demonstrated
- Those who reported increased brain activities such as reading, museum visits, puzzle playing, had higher test scores for memory and problem solving challenges

Lifestyle Changes

- Investigation into the effect of lifestyle changes on memory and thinking
- Interventions included nutritional guidance, physical exercise, cognitive training, social activities and CVD risk factor intervention
- Improvements seen in the intervention group

Thank You

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